Synthesis W. K. O. Teixeira et al. Review

Recent Advances in the Synthesis of Enantiomerically Enriched Diaryl, Aryl Heteroaryl, and Diheteroaryl Alcohols through Addition of Organometallic Reagents to Carbonyl Compounds

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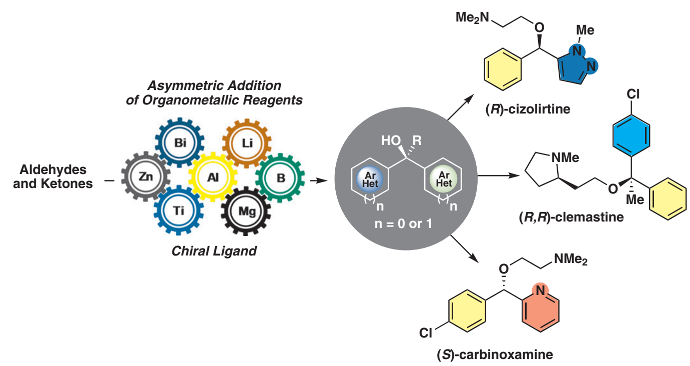

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Abstract Enantiomerically enriched diaryl, aryl heteroaryl, and diheteroaryl alcohols are an important family of compounds known for their biological properties. Moreover, these molecules are highly privileged scaffolds used as building blocks for the synthesis of pharmaceutically relevant products. This short review provides background on the enantioselective arylation and heteroarylation of carbonyl compounds, as well as, the most significant improvements in this field with special emphasis on the application of organometallic reagents.

Key words chiral alcohols, organometallic reagents, asymmetric addition, enantioselective arylation, heteroarylation

1 Introduction

Chiral alcohols are essential constituents of biologically active compounds and essential precursors in the synthesis of numerous useful derivatives. Among these, enantiomerically pure diaryl, aryl heteroaryl, and diheteroaryl alcohols are highly privileged scaffolds used as building blocks in organic synthesis; they have been used as precursors for the preparation of a number of biologically active compounds and pharmaceutically active substances, namely local anesthetics, antiarrhythmics, antidepressants, diuretics, laxatives, antispasmodic, antipruritic, antiparkinsonian, and anticholinergics.[1] Some important representative examples include (*S*)-carbinoxamine (**I**),[2] (*R*)-orphenadrine (**II**),[3] (*R*)-neobenodine (**III**),[4] and (*R*,*R*)-clemastine **IV**,[5] which all display antihistaminic activity (Figure 1). Cizolirtine (**V**) is an analgesic drug for patients with urinary incontinence.[6] Furthermore, chiral dithienylmethanols such as **VI** have showed relevant antiallergic and antiischemic activity.[7] Finally, LY2300559 (**VII**) has been investigated for the treatment of migraine with an oral absorption performance[8] and (*S*)-BMS184394 (**VIII**) displays activity against breast and leukemia cancer cells.[9]

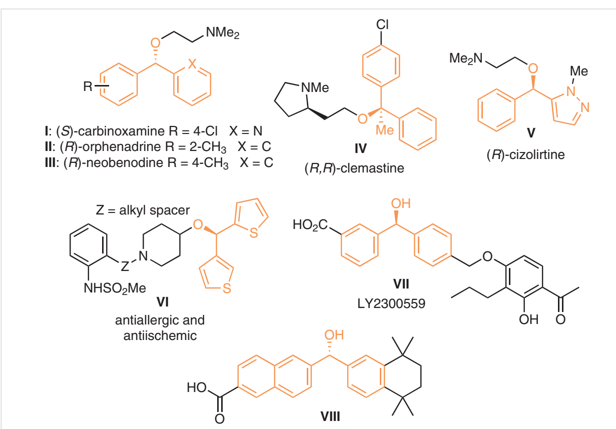

Figure 1 Examples of biologically active diaryl, aryl heteroaryl, and diheteroaryl alcohol derivatives

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Due to the importance of diaryl, aryl heteroaryl, and diheteroaryl alcohols, the enantioselective synthesis of this class of compounds has been the focus of many studies. In general, two synthetic approaches are available for their synthesis: the first strategy involves the enantioselective reduction of prochiral, diaryl or diheteroaryl ketones,[10] while the most common strategy comprises the asymmetric addition of aryl- and heteroarylmetal reagents to an aromatic or heteroaromatic carbonyl compound in the presence of a chiral ligand.[1,11] However, the choice of the organometallic reagent to be used is of crucial importance to maximize the selectivity during the enantioselective addition to the carbonyl compound as well as to avoid side reactions. Other important aspects to be considered are the commercial availability and the cost of these reagents. Although, some of them are sold by commercial suppliers, in

Biographical Sketches

Wystan Kreisly Othon Teixeira was born in Catanduva, SP, Brazil, in 1990. He studied chemistry at the Federal University of Rio Grande (Brazil) and obtained his Masters degree in 2017 from the same university. Since 2018 he has been a Ph.D. student in chemistry at the Federal University of São Carlos (UFSCar), under the guidance of Prof. Dr. Ricardo S. Schwab, working in the area of organic chemistry.

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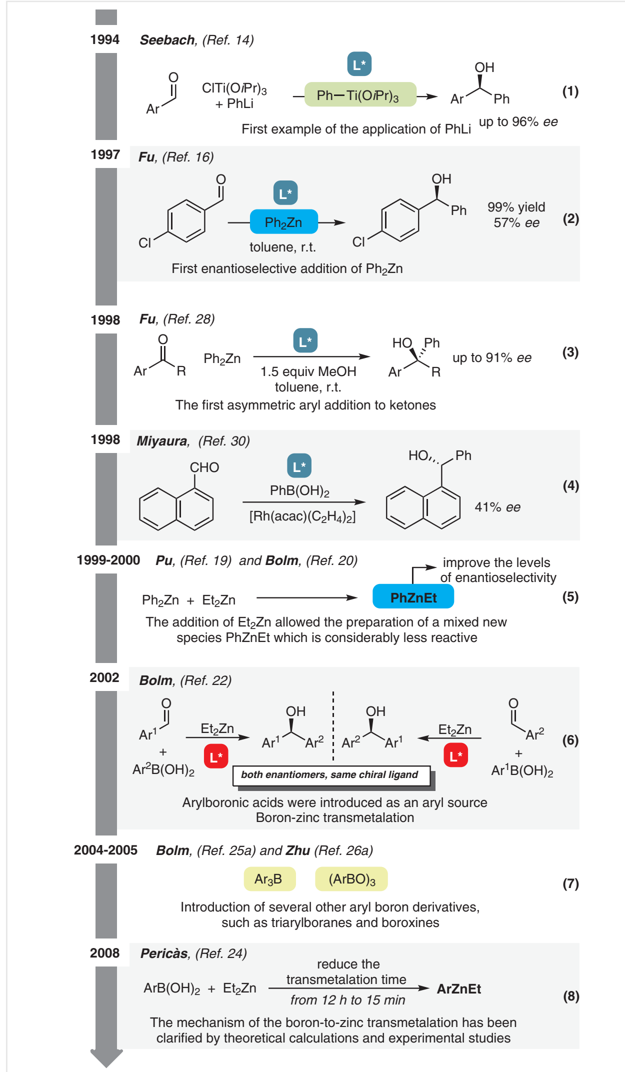
Ricardo S. Schwab was born in Santa Cruz do Sul (Brazil) in 1982. He received his B.S. in Industrial Chemistry in 2005 from the University of Santa Cruz do Sul (UNISC-Brazil). He obtained his Ph.D. in 2010 under the supervision of Prof. Dr. Antonio L. Braga (Federal University of Santa Maria, UFSM-Brazil) and co-supervision of Prof. Dr. Ludger Wessjohann [Leibniz Institute of Plant Biochemistry (IPB), Halle (Saale), Germany], working with catalytic enantioselective methodologies and on the synthesis of cyclopeptoids containing chalcogen mimetics. After a post-doctoral stay at the Federal University of Rio Grande do Sul (UFRGS-Brazil) in the group of Prof. Dr. Paulo H. Schneider, he became Assistant Professor in 2013 at the Federal University of São Carlos (UFSCar-Brazil). His research interest focuses on the development of more sustainable catalytic methodologies and asymmetric catalysis.

many cases availability and price restrict their use. Considering this point, it is extremely important that new methodologies are developed that generate organometallic species in situ from readily available starting materials which can act as an aryl and heteroaryl source groups.

The reviews written over the past have mainly focused on the application of organozinc reagents for the synthesis of enantioenriched diaryl, aryl heteroaryl, and diheteroaryl alcohols.[1,11a,b] To the best of our knowledge there is just one review that covers different methods for their synthesis in a racemic way.[12] Over the past, several great breakthroughs have been made in this field. Now many other organometallic reagents, as well as starting materials, can be efficiently used which has led to excellent yields and enantioselectivities. In light of the importance of chiral alcohols, especially the ones derived from aromatic or heteroaromatic compounds, there is a need for a more general review that illustrates the latest developments in this field. In this review, we survey the progress achieved in the asymmetric synthesis of diaryl, aryl heteroaryl, and diheteroaryl alcohols through the arylation of aldehydes and ketones in a catalytic enantioselective manner based on the type of organometallic reagent used as the aryl and heteroaryl source.

2 Background on the Enantioselective Synthesis of Diaryl, Aryl Heteroaryl, and Diheteroaryl Alcohols

The initial approaches to access enantioenriched diaryl alcohols involved the enantioselective addition of aryl Grignard reagents to aromatic aldehydes. Although this strategy has allowed the synthesis of enantioenriched diarylmethanols, the highly reactive nature of the Grignard reagents requires the use of stoichiometric quantities of either additives or chiral ligand.[13] The first effort to do this, in a catalytic manner, dates back to 1994,[14] when Seebach and co-worker employed $PhTi(O\textit{i}Pr)_3$ in combination with TADDOL-based titanium catalysts (Scheme 1, eq. 1). The phenyltitanium reagent was prepared by the reaction of $ClTi(O\textit{i}Pr)_3$ and PhLi followed by removal of the precipitated lithium salts by centrifugation. It should be highlighted that the presence of Lewis acids, such as LiCl, significantly reduced the enantioselectivity as they promoted unwanted racemic background reactions. Despite these results, the direct asymmetric addition of organomagnesium and organolithium reagents to carbonyl compounds remains unexplored, due to the challenge of controlling their high reactivity. On the other hand, the high functional group tolerance of organozinc reagents, especially the dialkylzinc contributed to making these reagents useful for the synthesis of optically active alcohols.[15]



Scheme 1 Timeline of the development of enantioselective catalytic arylation of aldehydes and ketones

In 1997, Fu and co-workers reported the first enantioselective catalytic addition of isolated diphenylzinc to aldehydes in the presence of a catalytic amount of planar-chiral azaferrocene (Scheme 1, eq. 2).[16] Although the product was obtained with only 57% *ee*, the pioneering studies by Fu paved the way for many subsequent studies.

Shortly thereafter, Bolm[17] and other research groups[18] contributed to improving the methodology by developing other catalysts for the asymmetric phenylation of aldehydes. However, this seemingly intuitive approach encountered unavoidable difficulties associated with diphenylzinc as an aryl source, such as high cost, high reactivity, and competitive racemic background reactions, making it a challenging reagent for the direct arylation of aldehydes. A major step forward was made independently by the groups of Pu[19] and Bolm,[20] when Et_2Zn was introduced into the reaction media (Scheme 1, eq. 5). This discovery allowed for the first time a dramatic improvement the levels of enantioselectivity for a wide variety of aldehydes. Furthermore, this new protocol allowed a reduction in the amount of expensive Ph_2Zn required. Bolm and co-workers[20a] discovered that when Ph_2Zn and Et_2Zn are mixed in solution, a group exchange takes place and the equilibrium is completely shifted towards the mixed species PhZnEt. The mixed ethylphenylzinc species allows the complete transfer of the phenyl group to the aldehyde, while the ethyl group behaves as a non-transferable group.[21]

However, the main drawback of the approaches using organozinc reagents discussed so far is the very limited reagent scope. Indeed, only the phenyl ring can be transferred using diphenylzinc as a commercially available diarylzinc source. Thus, in spite of its attractiveness, this method cannot be considered as a general approach for the synthesis of diarylmethanols. In this sense in 2002, Bolm and Rudolph reported the first general procedure for the catalytic asymmetric arylation of aldehydes (Scheme 1, eq. 6).[22] In this protocol, arylboronic acids were used as the aryl source of the nucleophilic aryl species, by a boron-to-zinc transmetalation reaction with diethylzinc. Since several arylboronic acids are commercially available or can be easily prepared, this straightforward methodology has broadened the scope of the reaction. Another interesting feature of this reaction is that both enantiomers of a given product can be prepared by using the same chiral ligand, with an appropriate choice of the reaction partners: arylboronic acid and aldehyde.

The work of Bolm marked a clear turning point, with the advent of more robust methodology for the synthesis of diaryl- and aryl(heteroaryl)methanols.[22,23] After this seminal work, the approach was further improved by the Pericàs group, who demonstrated that the boron-zinc transmetalation time can be significantly reduced (Scheme 1, eq. 8).[24] Several aryl boron derivatives like triarylboranes[23a,25] and boroxines[26] were also successfully used in the asymmetric arylation of aldehydes after Bolm's discovery (Scheme 1, eq. 7), and several other chiral ligands have also been shown to be compatible with this general process.[27]

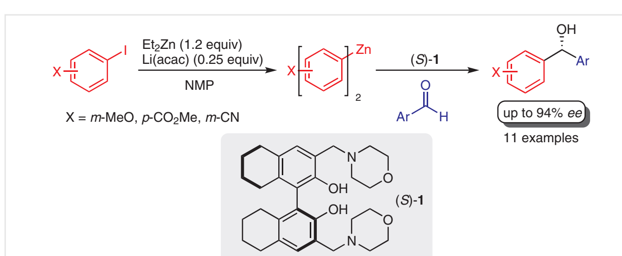
Despite promising advances in the catalytic asymmetric addition of organozinc to aldehydes for the synthesis of chiral diarylmethanols, the use of ketones for the synthesis of tertiary diaryl alcohols via catalytic asymmetric addition is still a formidable challenge, since the limitation comes from the inherent difficulty in differentiating the steric and electronic properties of both substituents in the prochiral ketone. Even with this limitation Fu and Dosa disclosed the first efficient asymmetric phenyl transfer to ketones in the presence of 15 mol% (+)-3-*exo*-(dimethylamino)isoborneol [(+)-DAIB] (Scheme 1, eq. 3).[28] This seminal work afforded tertiary alcohols with enantioselectivities up to 91% *ee*. Since this work there have been several reports concerning the preparation of chiral tertiary diaryl alcohols by arylzinc addition to ketones.[29]

An alternative approach for the synthesis of chiral diarylmethanols was proposed by Miyaura and co-workers in 1998,[30] who reported the application of an enantiopure monophosphine in the rhodium-catalyzed asymmetric addition of an arylboronic acid to 1-naphthaldehyde (Scheme 1, eq. 4). Although the product was obtained with only 41% *ee*, this work introduced a new way to achieve the synthesis of such alcohols. Recent advances in transition-metal-catalyzed arylations with arylboronic acids, such as the improvement in terms of catalyst efficiency, substrate scope, and reagents, have consolidated this protocol as an alternative to the traditional catalytic asymmetric addition of organometallic reagents [Ar_2Zn, Ar_3Al, $ArTi(O\textit{i}Pr)_3$, ArMgX and ArLi] to aldehydes or ketones.

3 Organozinc Reagents

To date, chiral diaryl alcohols are mainly obtained by the asymmetric addition of organozinc to carbonyl compounds. However, the catalytic enantioselective addition of functionalized arylzinc reagents to aldehydes was, until recently, problematic because of the difficulties in preparing functionalized arylzinc reagents, but since 2004 there have been impressive advances in this area.[31]

In 2004, the Knochel group developed the first synthesis of highly functionalized diarylzinc reagents. These reagents were prepared in situ from a Li(acac)-catalyzed iodine–zinc exchange reaction with Et_2Zn in the presence of *N*-methyl-2-pyrrolidone (NMP).[31a] This methodology was then used, in 2008, by Pu and co-workers for the synthesis of diarylmethanols.[32] The diarylzinc reagents, prepared by a modification of Knochel's method from aryl iodides and Et_2Zn, underwent enantioselective addition to aldehydes in the presence of a (*S*)-H_8-BINOL chiral ligand (*S*)-**1** (Scheme 2). The morpholino chiral alcohol **1**, derived from (*S*)-H_8-BINOL, has a dual function as it is responsible not only for high enantioselectivity for several substrates, but it also increases the nucleophilicity of the organozinc reagent. It is important to mention that individual optimization was required for each substituted aryl iodide (*m*-OMe, *p*-CO_2Me, and *m*-CN). The diarylmethanols were generally obtained in good yields and high enantioselectivities. The electronic



Scheme 2 Enantioselective arylation with functional arylzincs prepared from aryl iodides catalyzed by a (*S*)-H_8-BINOL-type chiral ligand

effects on the diarylzinc reagents played an important role, since diarylzinc reagents containing electron-donating groups required shorter reaction times.

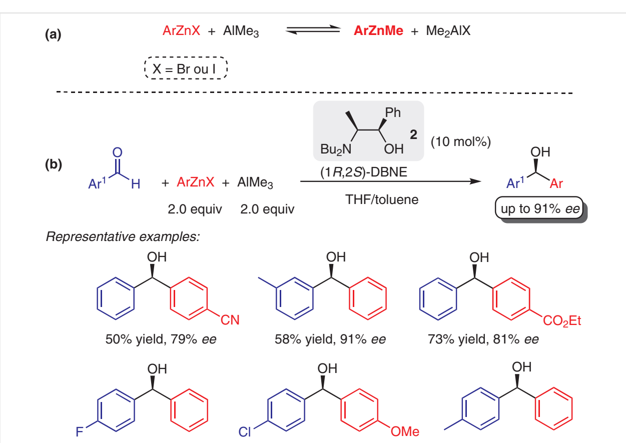
To extend the catalytic properties of the H_8-BINOL-type chiral ligands, Pu and co-workers prepared a series of H_8-BINOLs containing 3,3′-bis(tertiary amine) substituents.[33] This study showed that the H_8-BINOL-AM compounds with less sterically congested cyclic or acyclic tertiary amine substituents are more enantioselective than those with more bulky tertiary amine substituents. A good linear correlation of the enantioselectivity vs enantiomeric excess of the chiral ligand was observed, suggesting a monomeric active catalytic species. It was found for the first time that an aryl bromide, 2-bromothiophene, could be used to prepare a heteroarylzinc reagent by reaction with $ZnEt_2$. The addition of this heteroarylzinc reagent to an aldehyde proceeded with good enantioselectivity (86% *ee*).

Another catalytic system developed for the arylation of aldehydes using Knochel's[31a] method employed aziridine alcohols, which are readily accessible from chiral sources such as amino acids. Thus the enantioselective asymmetric additions of arylzinc species to aldehydes in the presence of 20 mol% of the chiral ligand afforded high yields and *ee's* of the arylmethanol products.[34]

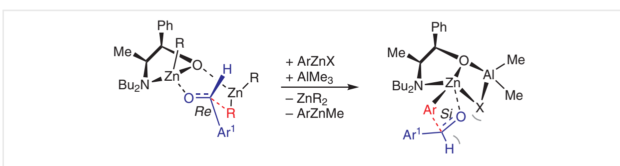
Organozinc halides (RZnX) have become an important class of reagents. They are especially useful because of their commercial availability and they can be readily used in the Pd- and Ni-catalyzed Negishi cross-coupling reaction.[35] Encouraged by the approach introduced by Bolm[20] in 2000, who found that the mixed reagent PhZnEt exhibited a slower background reaction than Ph_2Zn, in 2007 Woodward and co-workers applied ArZnX (X = Br or I) as the aryl source for the synthesis of diarylmethanols with functional groups on both aryl rings (Scheme 3).[36] However, the first attempt to promote asymmetric arylation of aldehydes using PhZnBr and R_2Zn (R = Me or Et) in the presence of (1*R*,2*S*)-(+)-dibutylnorephedrine [(1*R*,2*S*)-DBNE, **2**] afforded the product in poor yields only (<40%) and as near racemates.

Nonetheless, it was observed that the addition of trimethylaluminum to a commercial THF solution of ArZnX improved the results and simultaneously promoted the Schlenk equilibria to generate the reactive ArZnMe (Scheme 3a). Although the presence of $AlMe_3$ afforded the diarylmethanols with good enantioselectivities 79–91% *ee* (Scheme 3b), the facial selectivity is opposite to that expected for the classic ***anti*** transition state **A** originally described by Noyori for diethylzinc addition to aldehydes;[37] it was instead proposed that the reaction goes through transition state **B** (Scheme 4).[36,38] This model is supported by spectroscopic and mechanism studies. Investigations by ^{13}C NMR spectroscopy of mixtures of PhZnBr and $AlMe_3$ in THF indicated that a rapid ligand exchange takes place between zinc and aluminum. This result, associated with related experiments, allowed Woodward and co-workers to propose that $AlMe_3$ promotes the zinc Schlenk process generating a mixed organometallic species that leads to the formation of an atypical transition state, which is distinctly different from the classic '*anti*' transition state delineated by Noyori.[37]

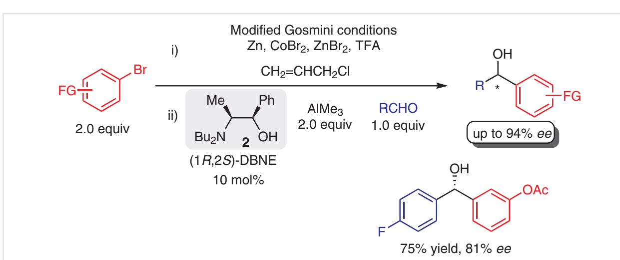
To improve the synthetic utility of this method, Woodward and co-workers extended the scope of the reaction to different aromatic aldehydes and functionalized arylzinc halides.[38] They were also able to generate ArZnBr in situ from aryl bromides and zinc dust by using a modified procedure developed by Gosmini and co-workers in 2003.[39] The ArZnBr generated in situ was directly used to arylated a range of aldehydes in the presence of 2 equivalents of $AlMe_3$ and 10 mol% of chiral ligand (1*R*,2*S*)-DBNE **2**



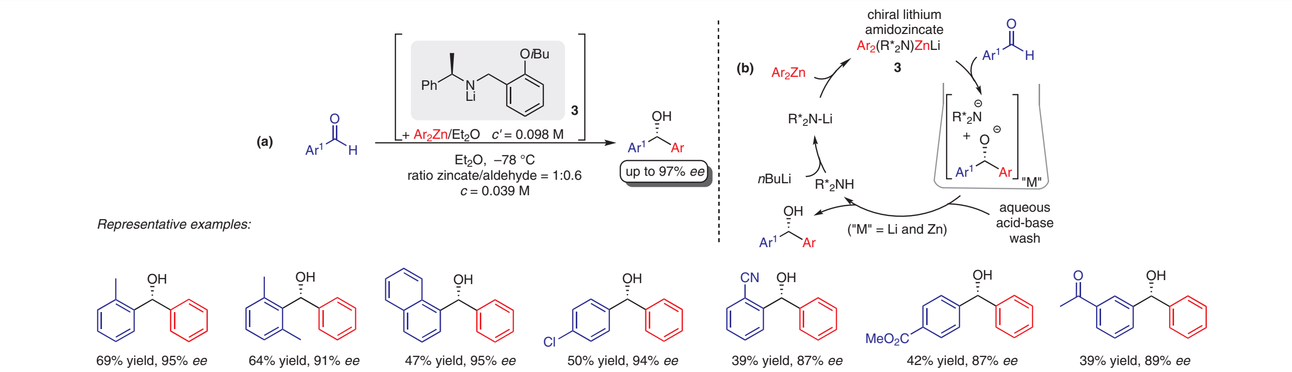
Scheme 3 (a) Aluminum-zinc Schlenk process; (b) asymmetric catalytic addition of ArZnX to aldehydes promoted by $AlMe_3$



Scheme 4 Transition states (**A**) proposed by Noyori and (**B**) proposed by the Woodward and co-workers



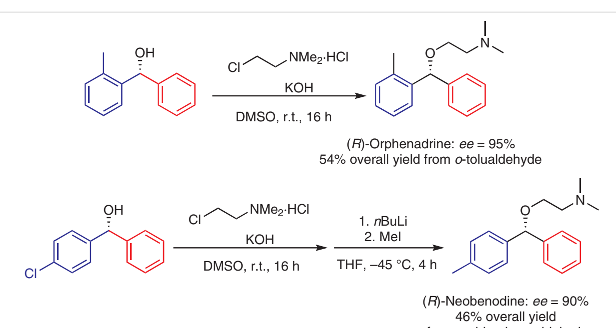
Scheme 5 Direct use of aryl bromides in preparation of secondary alcohols



Scheme 6 (a) Enantioselective nucleophilic 1,2-addition of chiral tricoordinated lithium amido arylzincate on aryl aldehydes. (b) Recycling of a chiral lithium amide during 1,2-nucleophilic additions of a chiral lithium amido zincate to aryl aldehydes.

(Scheme 5). The secondary alcohols were obtained with moderate to good enantioselectivities (30–94% *ee*) with highly reliable enantioface selectivity and functional group tolerance. However, just one example of the arylation of an aromatic aldehyde was reported using this method.

In 2019, the Harrison-Marchand group disclosed a new and efficient strategy for the enantioselective synthesis of diarylmethanols.[40] They utilized the concept of organo(bi)metallic 'ate-complex' species, which combine, in a synergic operation, specific properties of two organometallic compounds. As shown in Scheme 6a, the desired diarylmethanols were obtained in good yields and high enantiomeric excess for a range of aldehydes by using chiral tricoordinated lithium amido aryl zincates **3**. Good levels of induction were not restricted to *ortho*-substituted aromatic aldehydes, which was a major limitation of previous reported approaches. Moreover, one of the more relevant features of the chiral amido inducers is their ease of separation and reuse, by carrying out an acid-basic wash. The recovered ligand was reused at least five times with little or no loss of activity and enantioselectivity (Scheme 6b).



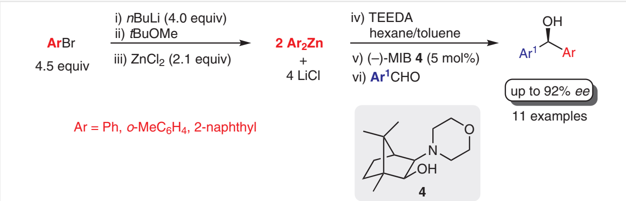
Scheme 7 Synthesis of (*R*)-orphenadrine and (*R*)-neobenodine

The new methodology was further utilized for the synthesis of (*R*)-orphenadrine, an anticholinergic drug for treating muscle pain and to help with motor control in Parkinson's disease, and (*R*)-neobenodine, which displays antihistaminic and anticholinergic activity. This methodology affords moderate yields (up to 54% overall yield) and excellent *ee's* (up to 95%) (Scheme 7).[40b]

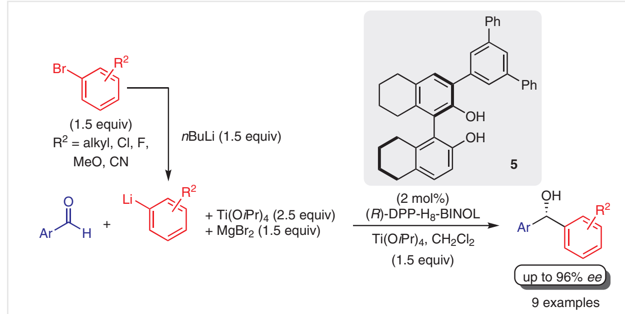
4 Organolithium Reagents

Organolithium reagents are the most common reagents used in organic synthesis and industrial applications.[41] However, for a long time, stoichiometric amounts of chiral ligands were used during the asymmetric addition of organolithium reagents to aldehydes due to the high reactivity of organolithium, and the products were mainly obtained with moderate to high enantiomeric excess.

After the pioneering work of Seebach and co-workers[14] in 1994, there was not significant improvement in the use of organolithium reagents as the source of aryl transfer group until the development of a new and efficient methodology in 2006 by Walsh and Kim for the asymmetric arylation of aldehydes with arylzinc reagents generated from the reaction of $ZnCl_2$ and aryllithium reagents.[42] The diorganozinc was prepared by reaction aryl bromide with *n*BuLi followed by lithium–zinc exchange reaction with $ZnCl_2$ in the presence of *t*BuOMe. *N*,*N*,*N'*,*N'*-Tetraethylethylenediamine (TEEDA) was then added to remove the LiCl byproduct from the reaction media, thus avoiding the need to separate the lithium salt from the reaction medium through centrifugation, followed by addition of (2*S*)-(–)-3-*exo*-morpholinoisoborneol [(–)-MIB **4**] and the aldehyde. Under these reaction conditions, the diarylmethanols were obtained with good to excellent enantioselectivities (80–92% *ee*) (Scheme 8).



Scheme 8 Enantioselective arylation of aldehydes by using organolithium as aryl source



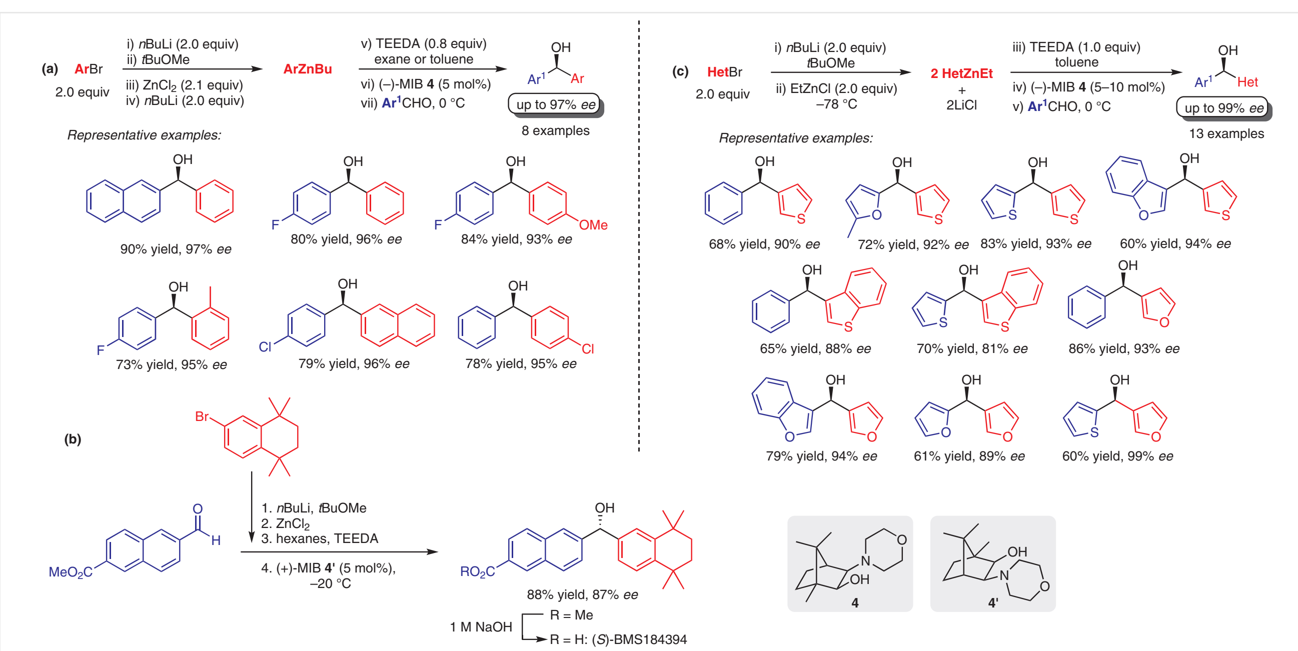
Scheme 10 Catalytic enantioselective synthesis of diarylmethanols from aryl bromides and aryl aldehydes by using organolithium reagents

Based on the strategy developed by Bolm[20] to circumvent the problem associated with the high reactivity of arylzinc, the Walsh group used a similar approach to generate aryl(butyl)zinc reagents (ArZnBu) from the treatment of Ar_2Zn with two equivalents of *n*BuLi. The new mixed organozinc reagent was used in situ in the asymmetric addition reaction to aryl aldehydes (Scheme 9a). This strategy improved the overall system performance since uncatalyzed background additions were suppressed. The methodology was used in the synthesis of a precursor to BMS184394, where the arylation product was obtained in 88% yield and 87% *ee* (Scheme 9b).

The new method was further applicable to heteroaryl bromides with some minor modifications to afford a range of synthetically important chiral aryl(heteroaryl)- and diheteroarylmethanols in good to high yields and enantioselectivities (81–99% *ee*, 60–86% yields) when added to aryl and heteroaryl aldehydes (Scheme 9c).[43]

An alternative protocol for enantioselective aryl transfer reaction to aldehydes starting from readily available aryl bromides was reported by the Harada group in 2010 (Scheme 10).[44] A variety of diarylmethanols were readily prepared in high yields and enantioselectivities by treatment of aryl bromides with *n*BuLi to prepare the corresponding aryllithium reagent. Then, the ArLi reagent was mixed with $Ti(OiPr)_4$ and $MgBr_2$ to generate a new mixed titanium reagent [ArLi-$MgBr_2$-$Ti(OiPr)_4$] that further undergoes enantioselective arylation of aromatic aldehydes in the presence of a titanium catalyst derived from 3-(3,5-diphenylphenyl)-H_8-BINOL (DPP-H_8-BINOL, **5**). This elegant approach require the used of only 2 mol% of chiral ligand. The higher enantioselectivity might be attributed to the presence of $MgBr_2$, because when it was not used the chiral diarylmethanols were obtained only in 50% *ee*.

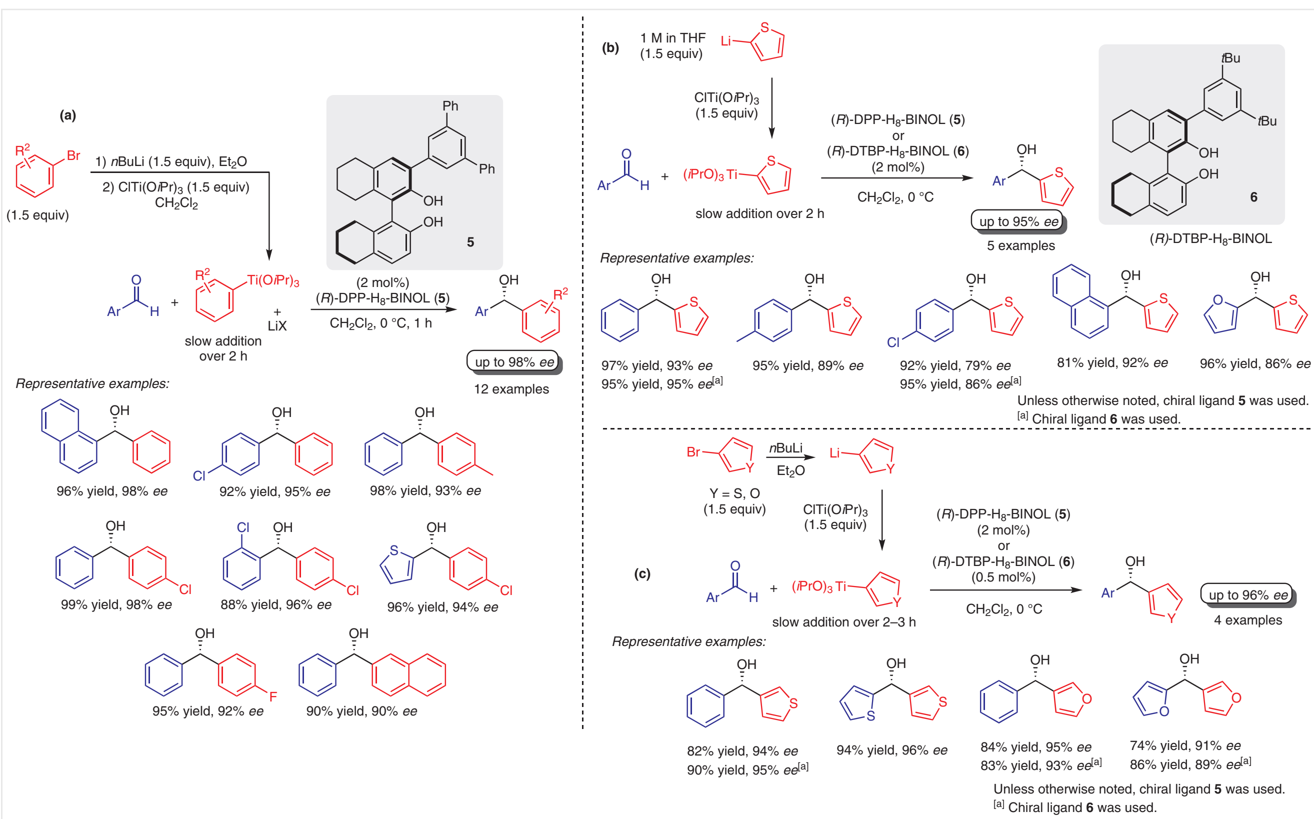
With the goal of developing a more practical, cost-effective, and scalable protocol for the enantioselective arylation



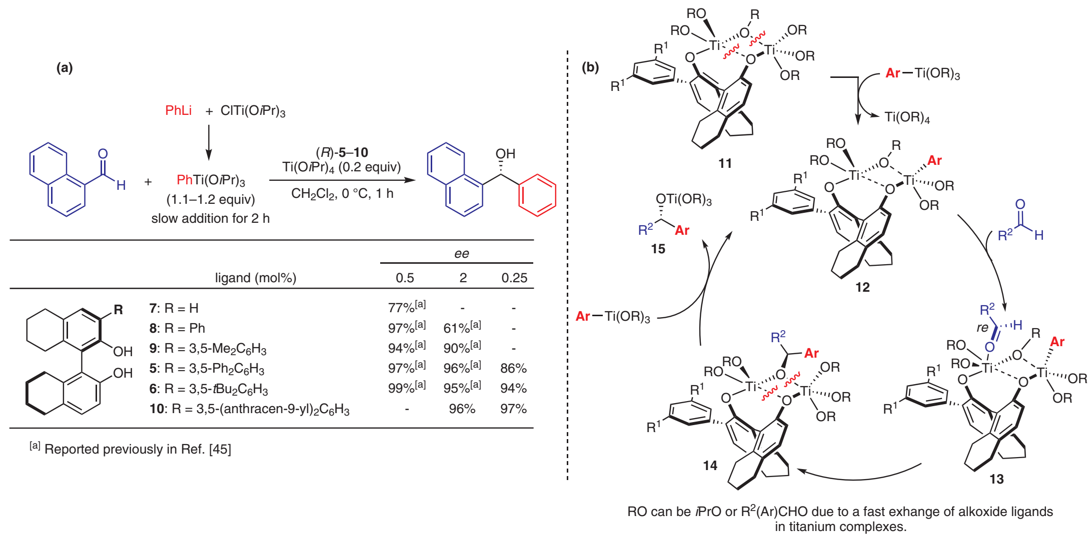
Scheme 9 (a) Catalytic asymmetric aryl additions to aldehydes with ArZnBr generated from aryl bromides. (b) Formal synthesis of (*S*)-BMS184394. (c) Synthesis of aryl(heteroaryl)- and diheteroarylmethanols.

and heteroarylation of aldehydes with organotitanium, the Harada group focused their attention on the transmetalation of aryllithium reagents with titanium to generate $ArTi(O\textit{i}Pr)_3$.[45] Initially the $PhTi(O\textit{i}Pr)_3$ was prepared by reaction of bromobenzene with *n*BuLi in Et_2O followed by subsequent transmetalation with $ClTi(O\textit{i}Pr)_3$ in CH_2Cl_2. Slow addition of the phenyltitanium reagent to a CH_2Cl_2 solution of 1-naphthaldehyde in the presence of 3-substituted H_8-BINOL-based chiral titanium complexes DPP-H_8-BINOL **5** or DTBP-H_8-BINOL **6** (DTBP = 3,5-di-*tert*-butylphenyl) was essentially to afford the enantioenriched secondary alcohol in high enantiomeric excess. It was not necessary to remove or inactivate the LiCl salt obtained during the transmetalation step and the enantioselectivity was not affected by this salt. The scope of the reaction was then extended to a series of aryl bromides and aryl aldehydes including sterically hindered examples, to afford the corresponding diarylmethanols in high yields and excellent enantiomeric excess (Scheme 11a). Subsequently, this procedure with some minor changes was applied successfully for the enantioselective heteroarylation of aldehydes by employing commercially available 2-thienyllithium (1 M in THF) (Scheme 11b). Moreover, the protocol could be extended to the enantioselective addition of 3-thienyl and 3-furyl groups to aldehydes; selected examples are shown in Scheme 11c. The reactions with aromatic and heteroaromatic aldehydes underwent heteroaryl addition with very high enantiomeric excesses and in very good yields. Remarkably, further reduction in the amount of catalyst did not result in an erosion of enantioselectivity. This methodology was performed on a 10 mmol scale using only 0.5 mol% of catalyst.

To clarify the origin of the activity and gain more insights into the reaction mechanism, the relation between the size of the substituents on the 3-position of H_8-BINOL and the activity of the resulting catalysts was investigated.[46] Six chiral ligands derived from (*R*)-H_8-BINOL **5**–**10** were evaluated (Scheme 12a). The presence of the substituent at the 3-position of these ligands had a beneficial effect on the enantioselective arylation of an aldehyde with $PhTi(O\textit{i}Pr)_3$ and, remarkably, the enantioselectivity was significantly enhanced by increasing the size of this substituent. When the catalyst loading was reduced, only ligands bearing a sterically demanding substituent at the 3-position afforded the product with high selectivity. It was proposed that the enhanced catalytic activity is associated with the acceleration of the catalyst turnover step due to the facilitated generation of the tetracoordinate bis-titanium intermediate by the cleavage of the weak intramolecular aggregation bonds as demonstrated by the X-ray and VT-NMR studies. A mechanism for the reaction has been proposed based on the data



Scheme 11 (a) Catalytic enantioselective arylation of aldehydes by using bromides as aryl source. (b) Catalytic enantioselective addition of the 2-thienyl group to aldehydes. (c) Enantioselective 3-thienylation and 3-furylation of aldehydes.



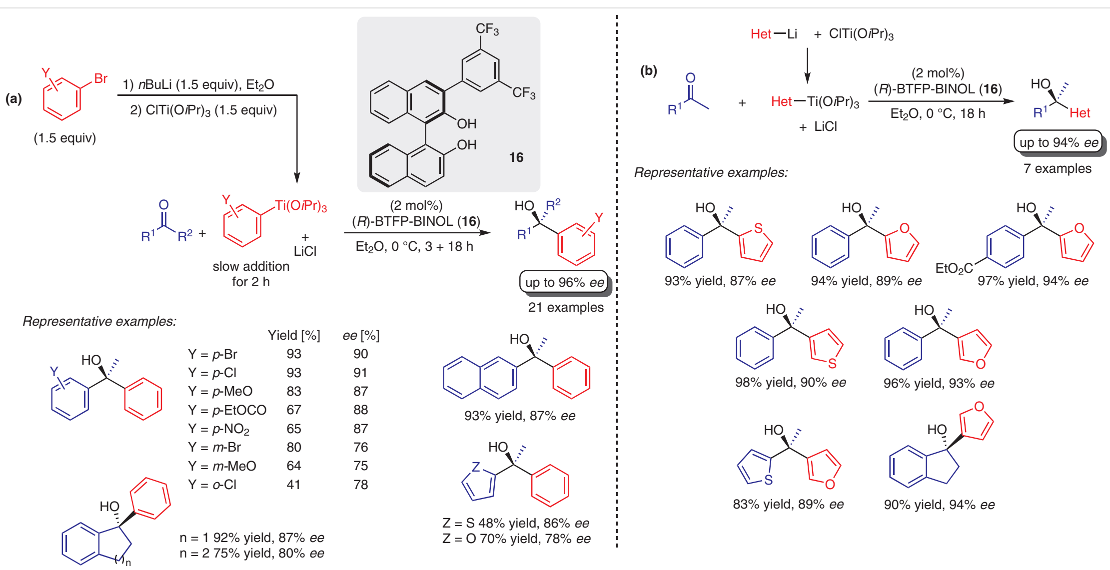
ligand (mol%)	ee 0.5	ee 2	ee 0.25
7: R = H	77%[a]	-	-
8: R = Ph	97%[a]	61%[a]	-
9: R = 3,5-Me$_2$C$_6$H$_3$	94%[a]	90%[a]	-
5: R = 3,5-Ph$_2$C$_6$H$_3$	97%[a]	96%[a]	86%
6: R = 3,5-*t*Bu$_2$C$_6$H$_3$	99%[a]	95%[a]	94%
10: R = 3,5-(anthracen-9-yl)$_2$C$_6$H$_3$	-	96%	97%

[a] Reported previously in Ref. [45]

Scheme 12 (a) Enantioselective addition of PhTi(O*i*Pr)$_3$ to 1-naphthaldehyde at various catalyst loadings. (b) Plausible catalytic cycle for titanium-catalyzed arylation of aldehydes.

collected. First, the aryl group of ArTi(O*i*Pr)$_3$ is transferred to complex **11** to generate complex **12** (Scheme 12b). The complex **12** undergoes coordination of aromatic aldehyde to generate the activated complex **13**. Subsequently, the enantioselective arylation takes place affording the intermediate **14** which then undergoes aryl group transfer with ArTi(O*i*Pr)$_3$ to provide the product titanium alkoxide **15** with simultaneous regenerating of complex **12**.

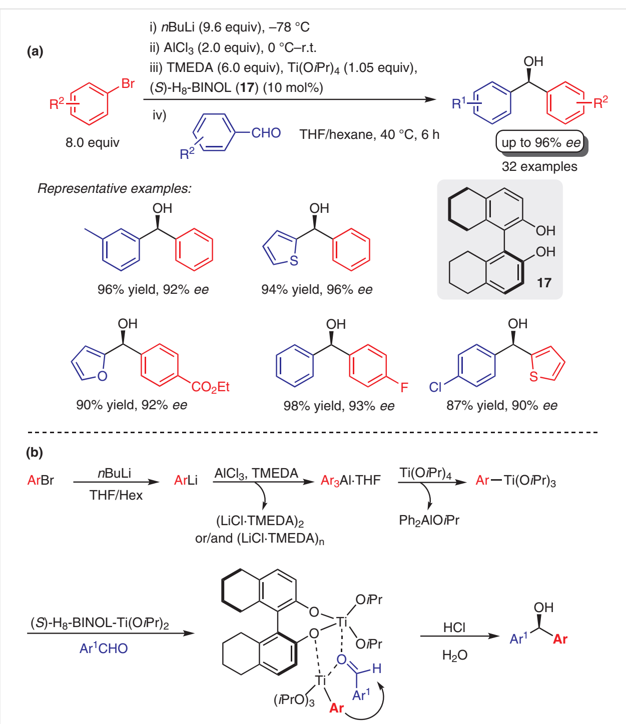
There are few examples of efficient and highly enantioselective addition of organometallic aryl reagents to ketones, and most approaches involve the use of the organozinc reagents.[29,47] In 2017, the Harada group reported the synthesis of tertiary diaryl- and aryl(heteroaryl)alcohols starting from commercially available low-cost aryl ketones and aryl or heteroaryl bromides (Scheme 13).[48] Thus, by employing a similar approach used for the arylation and heteroarylation of aromatic aldehydes, they were able to generate in situ the organotitanium reagents [ArTi(O*i*Pr)$_3$ and HetTi(O*i*Pr)$_3$] followed by enantioselective addition to ketones in the presence of a chiral titanium catalyst derived from (*R*)-3-[3,5-bis(trifluoromethyl)phenyl]-1,1′-bi-2-naphthol (BTFP-BINOL, **16**).



	Yield [%]	*ee* [%]
Y = *p*-Br	93	90
Y = *p*-Cl	93	91
Y = *p*-MeO	83	87
Y = *p*-EtOCO	67	88
Y = *p*-NO$_2$	65	87
Y = *m*-Br	80	76
Y = *m*-MeO	64	75
Y = *o*-Cl	41	78

Scheme 13 (a) Catalytic enantioselective preparation of tertiary alcohols starting from aryl bromides and ketones. (b) Catalytic enantioselective preparation of tertiary aryl(heteroaryl)alcohols.

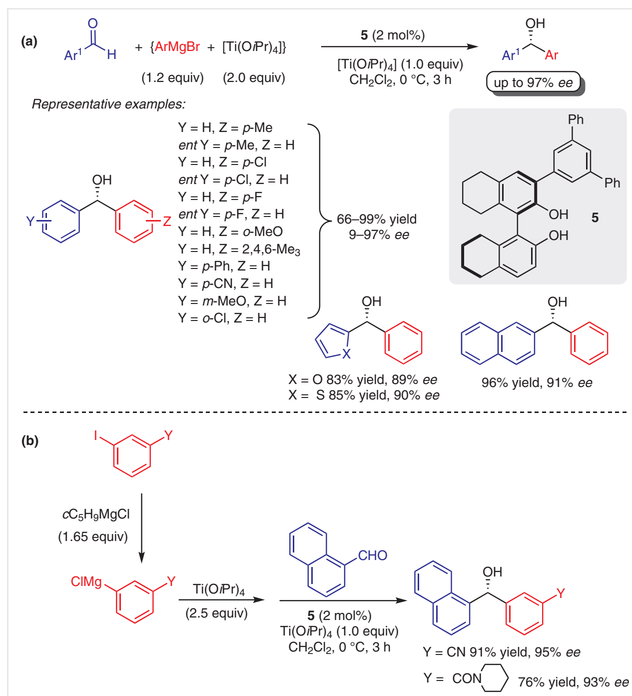
In 2014, the Da group reported an alternative methodology that allows the use of aryl bromides as inexpensive starting materials for the enantioselective arylation of aldehydes in the presence of (*S*)-H_8-BINOL (**17**) (Scheme 14a). This new strategy involves the in situ conversion of aryl bromides into aryltitaniums.[49] Remarkably high enantioselectivities (83–96% *ee*) and good yields (71–96%) were obtained for a variety of aromatic as well as heteroaromatic aldehydes. Overall, this methodology adds to the chemists' toolbox for the enantioselective arylation of aryl aldehydes. It can be regarded as a more efficient synthesis as the protocol does not require salt-free aryltitanium intermediates, slow addition of aryltitanium to aldehydes over a long period, or a change of the solvent during the reaction process. The proposed mechanism is shown in Scheme 14b. Operationally simple, aryllithiums were first prepared at –78 °C from the reaction of aryl bromides and *n*BuLi. The addition of $AlCl_3$ allows aryl transfer to the triarylaluminum and generates the achiral Lewis acid LiCl. The aryl group is finally transferred to titanium by treating the Ar_3Al with $Ti(OiPr)_4$. The resulting $ArTi(OiPr)_3$ delivers the aryl group to the aldehyde catalyzed by (*S*)-H_8-BINOL-$Ti(OiPr)_2$.



Scheme 14 (a) Catalytic asymmetric arylation of aldehydes. (b) Proposed reaction mechanism.

5 Grignard Reagents

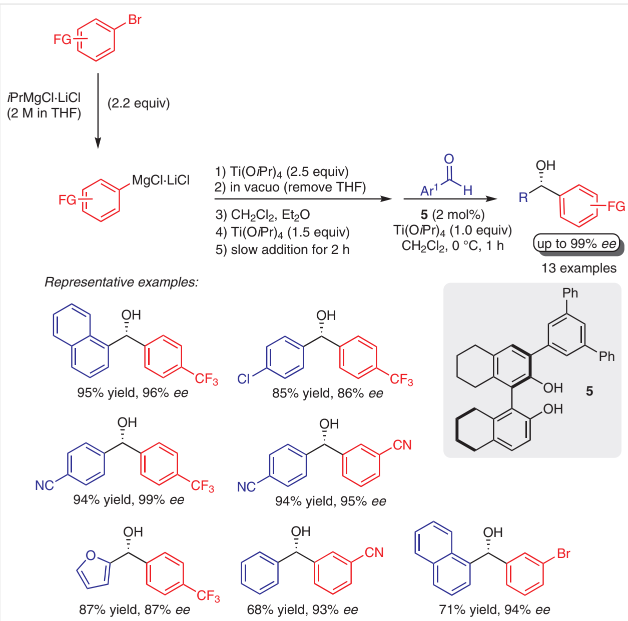
The catalytic asymmetric addition of aryl Grignard reagents to aldehydes for the synthesis of diarylmethanols remained unexplored until recently because of their high reactivity.[50] The first step towards this was taken in 2008 by the Harada group who developed the first methodology for the enantioselective arylation of aldehydes by using an almost stoichiometric amount of the arylmagnesium reagents in combination with excess $Ti(OiPr_4)$ in the presence of a chiral catalyst derived from **5** and $Ti(OiPr)_4$ (Scheme 15a).[51] In general, the diarylmethanols were obtained in excellent yields and high enantioselectivities. Inspired by the work of the Knochel group[52] they used low temperatures for the preparation of thermally unstable functionalized Grignard complexes from halogen–magnesium exchange of the corresponding haloarenes and reported a practical and mild protocol for the enantioselective synthesis of highly functionalized aryl secondary alcohols. The method is applicable to aryl iodides bearing a cyano group or an amide group (Scheme 15b). This approach expands significantly the scope of the reaction and can be considered an interesting alternative to the use of functionalized commercially available arylboronic acids for the synthesis of highly functionalized diarylmethanols, which are very important precursors to many biologically active compounds.



Scheme 15 (a) Catalytic asymmetric arylation of aldehydes with Grignard reagents. (b) Catalytic asymmetric transfer of functionalized aryl groups with Grignard reagents.

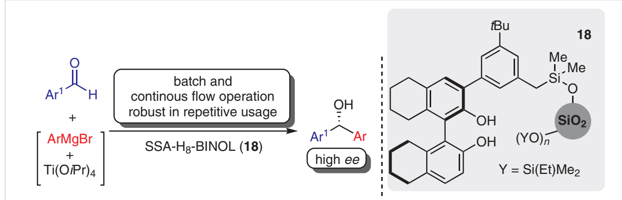
In 2010, the Harada group extended the application of this catalytic system to other functionalized Grignard reagents serving as the aryl source.[53] Grignard reagents bearing functionalized groups (e.g., ester/amide, carbonyls, cyano, and bromo groups) were compatible with the reaction conditions and successfully afforded diarylmethanols with high *ee* and yields.

Although the previous protocol developed by the Harada group has attracted considerable attention, its application to functionalized aryl bromides remains limited. In 2011, the Harada group made an interesting contribution to this field,[54] which utilized the efficient, catalytic asymmetric aryl transfer reaction to aldehydes using the titanium complex of (*R*)-DPP-H_8-BINOL (**5**) as the catalyst and functionalized arylmagnesium species that were prepared in situ by treatment of the less expensive aryl bromides with *i*-PrMgCl·LiCl (Scheme 16).[55] The method was also applicable to aryl bromides bearing CF_3, Br, or CN groups at the *meta* and *para* positions, affording enantiomerically enriched functionalized diarylmethanols at low catalyst loading (2 mol%).



Scheme 16 Catalytic enantioselective arylation of aldehydes by using functionalized Grignard reagents generated from aryl bromides

From a sustainable chemistry point of view, recycling and reuse of catalysts is an important aspect. Different types of materials have been widely used as insoluble supports for the attachment of chiral ligands. In 2017, the Harada group developed a synthetic procedure for the attachment of 3-aryl-H_8-BINOL to the surface of silica gel using a hydrosilane derivative as a precursor. The silica-supported 3-aryl-H_8-BINOL ligand **18** was used to mediate the reaction of Grignard reagents in the presence of Ti(O*i*Pr)$_4$ with aromatic aldehydes (Scheme 17).[56] The recovered ligand was reused at least ten times and maintained high enantioselectivities (88–95% *ee*) and yields (85–99%). Moreover, the practicality of this heterogeneous catalyst system was further demonstrated by recycling experiments in which the same immobilized catalyst was used with different combinations of aldehydes and Grignard reagents.

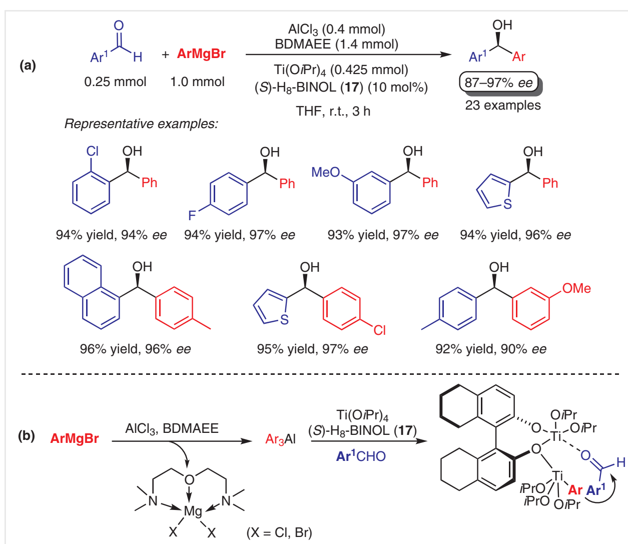


Scheme 17 Silica-supported catalyst for enantioselective arylation of aldehydes under batch and continuous-flow conditions

The silica-supported 3-aryl-H_8-BINOL chiral ligand **18** was also used for the asymmetric synthesis of diarylmethanols by the enantioselective arylation of aromatic aldehydes using organomagnesium reagents under continuous-flow conditions (Scheme 17).[57] A simple pipet reactor packed with the heterogeneous ligand allowed the production of diarylmethanols with good to excellent enantioselectivity (81–96%) through the introduction of aldehydes and mixed titanium reagents generated from the organomagnesium precursors. Moreover, this system could be used repeatedly in different reactions without appreciable deterioration of the activity.

In 2014, Yus, Maciá, and Fernández-Mateos used, 1-naphthyl-BINMOL ligand and excess Ti(O*i*Pr$_4$) (10 equiv) for the asymmetric addition of aryl Grignard reagents to ketones. In most cases, the tertiary alcohols were obtained with moderate to good enantioselectivities (46–92% *ee*).[58]

The Da group reported, in 2010, a practical catalytic asymmetric arylation of aldehydes by using a variety of aryl and heteroaryl Grignard reagents (Scheme 18).[59] Grignard reagents were used as an aryl source and according to the procedure, they are first converted into the corresponding triarylaluminum by reaction with $AlCl_3$. During the reaction



Scheme 18 (a) Catalytic asymmetric ArMgBr addition to aldehydes. (b) Mechanism for the (*S*)-H_8-BINOL catalyzed addition of ArMgBr to aldehydes in the presence of $AlCl_3$ and BDMAEE.

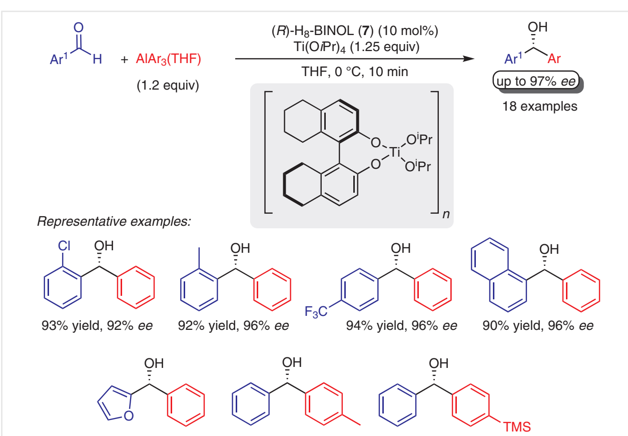
magnesium salt byproducts are generated that can readily promote the background reaction to form a racemic product. To overcome this problem bis[2-(dimethylamino)ethyl] ether (BDMAEE) was used as an additive to chelate the magnesium salts. Then Ar_3Al transfers an aryl group to $Ti(O\textit{i}Pr)_4$ generating $ArTi(O\textit{i}Pr)_3$, this forms a bimetallic nuclear complex with (*S*)-H_8-BINOL-$Ti(O\textit{i}Pr)_2$ and aldehyde that affords the product with excellent enantioselectivities (≥98%). This protocol allowed the synthesis of diarylmethanols at room temperature with inexpensive Grignard reagents thus facilitating the scale up of the reaction.

6 Organoaluminum Reagents

In recent years, organoaluminum reagents have been effectively used for asymmetric additions to carbonyl compounds because they are more reactive than the organozinc or organoboron reagents.[60]

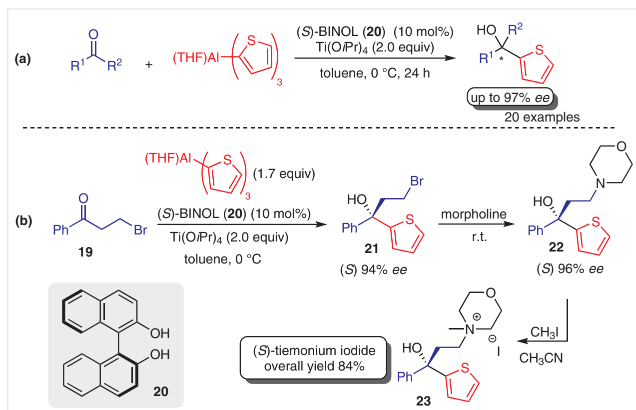
In 2006, the Gau group[61] described the first asymmetric aryl addition of an organoaluminum reagent, triaryl(tetrahydrofuran)aluminum [$AlAr_3$(THF)], to an aldehyde catalyzed by the titanium(IV) complex of (*R*)-H_8-BINOL (**7**). Using this reagent, aryl and heteroaryl aldehydes underwent arylation using 10 mol% of the catalyst with 10 min reaction time to give the corresponding diarylmethanols in greater than 97% *ee* (Scheme 19). The suppression of the background reaction is not required.

The application of [$AlAr_3$(THF)] reagents was further extended for the synthesis of tertiary diarylalcohols in high yields and excellent enantioselectivities, through the asymmetric aryl addition to ketones catalyzed by the titanium(IV) catalyst of (*S*)-BINOL.[62]



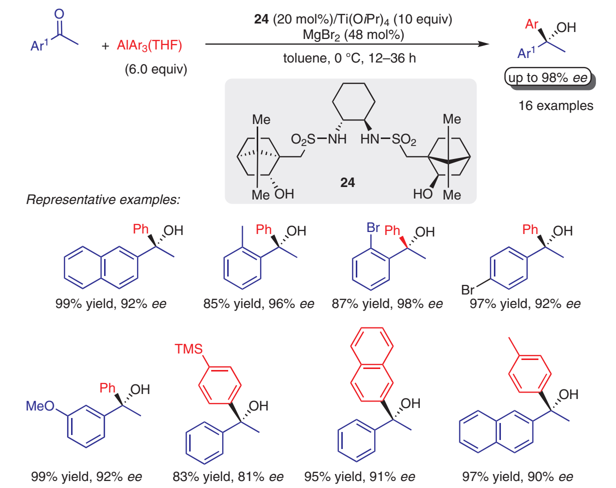
Scheme 19 Enantioselective titanium(IV)–(*R*)-H_8-BINOLate catalyst for arylation to aldehydes by [$AlAr_3$(THF)] reagents

In a further extension of this study, the Gau group developed the asymmetric catalytic 2-thienylaluminum[63] addition to a variety of aromatic ketones (Scheme 20a). This organoaluminum reagent allowed the synthesis of chiral 1-aryl-1-(2-thienyl)ethanols, which are bioactive compounds as well as useful synthetic intermediates. The catalytic system works excellently for aromatic ketones, having either an electron-donating or -withdrawing substituent on the aromatic ring. The 2-thienylaluminum reagent Al(2-thienyl)$_3$(THF) was used for the synthesis of chiral tiemonium iodide an anticholinergic/spasmolytic drug.[64] The key step was the synthesis of intermediate **21** by asymmetric catalytic 2-thienyl addition to 3-bromo-1-phenylpropan-1-one (**19**) catalyzed by a titanium catalyst of (*S*)-BINOL **20**, and this was followed by the treatment of **21** with morpholine to produce **22** in high yield (Scheme 20b). Compound **22** reacted with methyl iodide to furnish tiemonium iodide (**23**). The X-ray diffraction of a single crystal confirmed the *S*-configuration of the final product. This three-step synthesis produced (*S*)-tiemonium iodide in 84% overall yield.

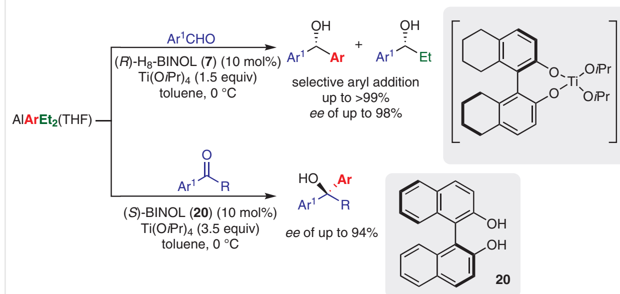


Scheme 20 (a) Chiral aryl 2-thienyl alcohols synthesized by 2-thienylaluminum transfer to aromatic ketones. (b) Asymmetric synthesis of (*S*)-tiemonium iodide.

The chiral complex generated from dihydroxy bis(sulfonamide) ligand **24** and $Ti(O\textit{i}Pr_4)$, previously successfully used for phenylzinc[29a,b] and arylzinc[29c] addition to aldehydes, was used by the Gau group for the asymmetric $AlAr_3$(THF) addition to ketones.[65] It was found that magnesium halide in an impure sample of $AlAr_3$(THF) plays a key role in the promotion of the asymmetric catalytic reaction and improves both the yields and enantioselectivity. The effect of a variety of additives was examined to better understand the key role of the metal salts in the promotion of the reaction, and it was found that 48 mol% of $MgBr_2$ afforded the product with better results. Selected examples are shown in Scheme 21. The methodology works well for a range of aryl methyl ketones possessing either electron-donating or electron-withdrawing groups with high levels of enantioselectivity. With regard to steric effects, it was observed that steric hindrance does play an important role since longer reaction times were required for *ortho*-substituted aromatic ketones to give the expected 1,1-diarylethanol derivatives in good yields.



Scheme 21 Asymmetric aryl addition of $AlAr_3$(THF) to ketones catalyzed by the titanium(IV) catalytic system promoted by $MgBr_2$



Scheme 23 Enantioselective arylation of aldehydes and ketones using $AlArEt_2$(THF) as aryl sources

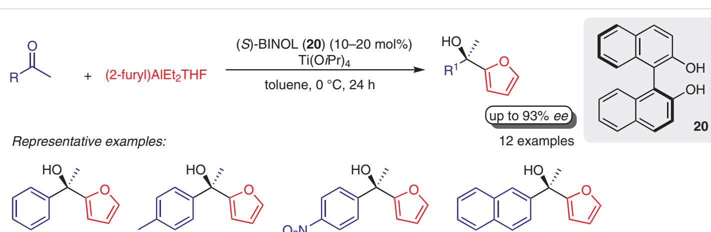
In subsequent work in 2008, the Gau group reported a new procedure for the asymmetric 2-furyl addition of (2-furyl)$AlEl_2$(THF) to aromatic ketones catalyzed by the titanium catalyst of 10–20 mol% (*S*)-BINOL (**20**) (Scheme 22).[66] The scope of the furylaluminum transfer to other ketones was examined and 1-aryl-1-(2-furyl)ethanols were obtained with 87–93% ee. The (2-furyl)$AlEl_2$(THF) addition to ketones occurred with higher selectivity for furyl transfer, since no ethylation products were observed.

To improve the atomic efficiency of arylaluminum reagents for asymmetric catalytic aryl additions to carbonyl compounds, in 2009 the Gau group reported a highly enantioselective arylation of aldehydes and ketones using $AlArEt_2$(THF) as the aryl source (Scheme 23).[67] The arylaluminum reagent was prepared by reaction of the corresponding aryl Grignard reagent with $AlEt_2Br$(THF) in THF. Application of various aldehydes in the aryl transfer reaction using a titanium(IV) complex of (*R*)-H_8-BINOL (**7**) allowed the synthesis of optically active diarylmethanols with up to 98% *ee*. For some aldehydes the ethyl group also competed for the addition to give undesirable ethylation products in 8–13% yields. This observation is in accordance with previous results and can be explained based on the fact that the transfer of sp^2-hybridized carbon-based substituents of aluminum reagents is more facile than that of sp^3-hybridized carbon-based substituents. On the other hand, the catalytic asymmetric aryl addition of $AlArEt_2$(THF) to ketones employing a titanium(IV) catalyst of (*S*)-BINOL **20** afforded the expected 1,1-diarylethanols as the sole products with excellent enantioselectivities.

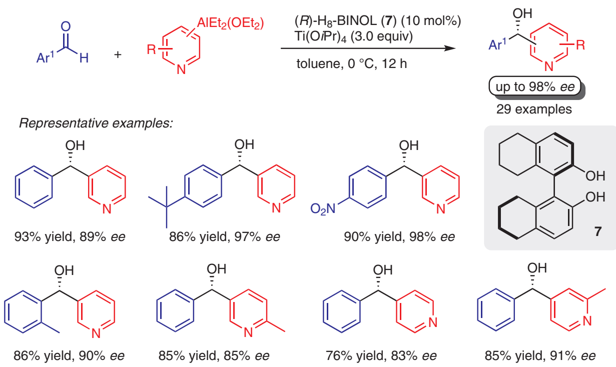
In 2015, Zhang and co-workers developed effective organoaluminum reagent for enantioselective pyridyl transfer reaction to aldehydes in the presence of an (*R*)-H_8-BINOLate titanium catalyst (Scheme 24).[68] The pyridyl aluminum reagents (pyridyl)$AlEt_2(OEt_2)$ were easily prepared by the reaction of $AlEt_2Cl$ with the corresponding pyridyllithiums produced in situ from the halogen–lithium exchange of pyridyl bromide with *n*BuLi. The catalytic system afforded a series of chiral aryl(heteroaryl)methanols containing various pyridyl groups in high yields with excellent enantioselectivities. Interestingly, different from the addition of $AlPhEt_2$(THF) to aldehydes,[67] the addition of (3-pyridyl)$AlEt_2(OEt_2)$ afforded the pyridyl product exclusively.

In subsequent work in 2018, Zhang and co-workers reported the catalytic asymmetric addition of 3-thienylaluminum reagent (3-thienyl)$AlEt_2(OEt_2)$ to ketones catalyzed by the titanium–(*S*)-BINOL catalytical system for the synthesis of optically active 1-aryl-1-(thienyl)ethanols.[69]

A major improvement was made in 2017 by the Da group. *N*-Methylmorpholine (NMM) was used as an effective additive in the synthesis of chiral diarylmethanols through the enantioselective arylation of aryl aldehydes catalyzed by (*S*)-H_8-BINOL–Ti(O*i*Pr)$_2$ (Scheme 25).[70] The nucleophilic arylaluminum reagent was in situ prepared from

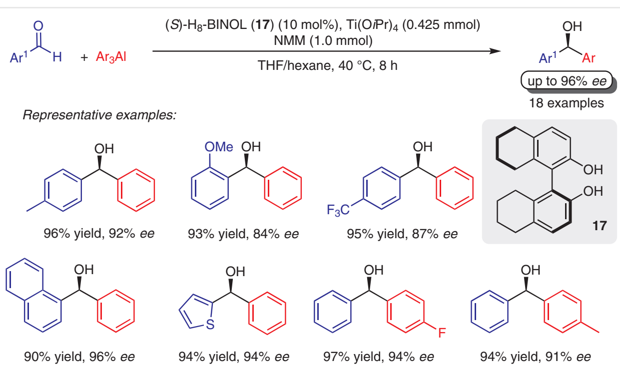


Scheme 22 $AlEt_2$(2-furyl)(THF) additions to ketones catalyzed by the in situ formed titanium complex of (*S*)-BINOL



Scheme 24 Enantioselective addition of pyridylaluminum reagents to aryl aldehydes

aryl bromide, *n*BuLi, and anhydrous $AlCl_3$ in THF. This work allowed, for the first time, a reduction in the loading of the triarylaluminum reagent by 50% from two equivalents to equimolar with respect to aldehydes, without compromising the yield and enantioselectivity of the process. The less expensive additive *N*-methylmorpholine (NMM) efficiently replaced the commonly used bis[2-(dimethylamino)ethyl] ether (BDMAEE) or *N*,*N*,*N*′,*N*′-tetramethylenediamine (TMEDA) additives that are usually used to inhibit the racemic background reactions.

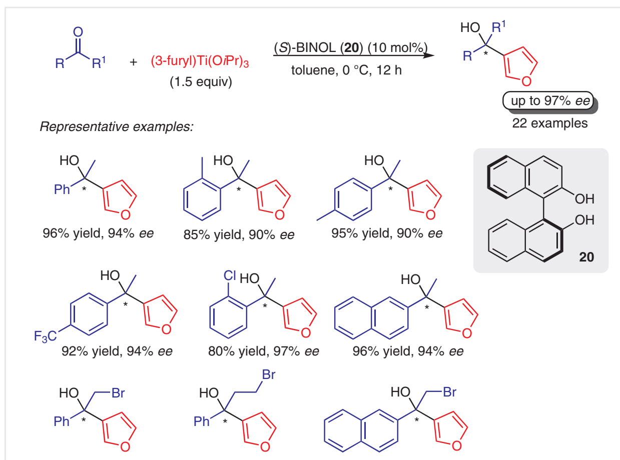


Scheme 25 Catalytic asymmetric arylation of aryl aldehydes

7 Organotitanium Reagents

Over the past few decades, enantioselective titanium-mediated transformations have been widely used to access diverse scaffolds of both synthetic and biological interest.[71] In this context, organotitanium reagents, more specifically $Ti(OiPr)_4$, have been extensively used to promote the enantioselective addition of the most commonly used organometallic compounds, such as, Grignard (RMgX, see Section 5), organozinc (R_2Zn, see Section 3), and organoaluminum (R_3Al, see Section 6) reagents, to carbonyl compounds. In general, a high excess of the $Ti(OiPr)_4$ is required in order to allow the transmetalation of an organic compound from the organometallic reagent (Mg, Zn, or Al) to form the organotitanium reagent, as well as, for the formation of the chiral titanium catalyst.[11b,c,72] However, the direct asymmetric addition of an aryltitanium reagent to a carbonyl compound is much less common.

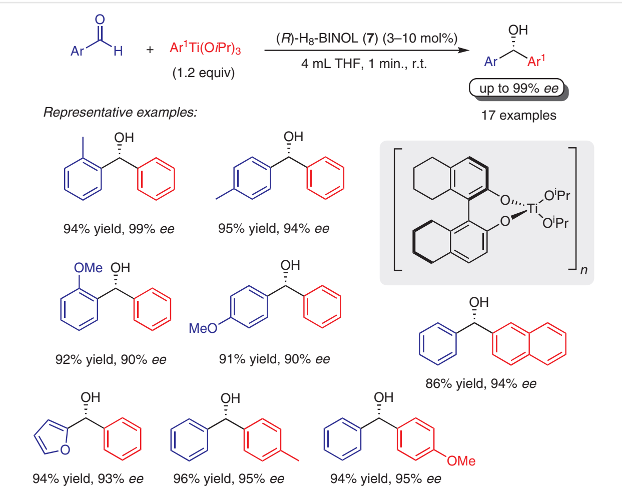
Among the pioneers in the field of titanium-catalyzed enantioselective arylation of carbonyl compounds, the Gau group reported a direct asymmetric catalytic addition of an organotitanium reagent in the presence of a chiral ligand (Scheme 26).[73] The (3-furyl)titanium reagent [(3-furyl)-$Ti(OiPr)_3]_2$ was successfully prepared by the reaction of 3-furyllithium with $Ti(OiPr)_3Cl$ in THF at –78 C. Based on an X-ray diffraction study of this complex, it was suggested that the (3-furyl)tri(isopropoxy)titanium complex has a dimeric structure with a Ti_2O_2 core bridging through the oxygen atom of two isopropoxy ligands. For the sake of simplicity the complex is represented by (3-furyl)$Ti(OiPr)_3$. Using as a standard ketone acetophenone with 10 mol% of (*S*)-BINOL (**20**) and 1.5 equivalents of (3-furyl)$Ti(OiPr)_3$ at 0 °C, 1-(3-furyl)-1-phenylethanol was obtained in a 100% conversion with an excellent 94% *ee*. Low conversion (30%) was observed when (3-furyl)$Ti(OiPr)_3$ was prepared in situ. The scope of the reaction was investigated with regard to the aryl ketone precursor. A wide variety of substituted aryl ketones, α- or β-halophenones and acetylfuran were examined and afforded 1-aryl-1-heteroarylalkan-1-ols in good to excellent yields with ≥90% *ee* (Scheme 26). Additionally, it was demonstrated that the reaction does not require the addition of $Ti(OiPr)_4$.



Scheme 26 Enantioselective 3-furyl addition to ketones catalyzed by the titanium catalyst of (*S*)-BINOL

In addition, the Gau group also reported the direct enantioselective arylation of aldehydes with $ArTi(OiPr)_3$ at room temperature in the presence of a chiral ligand.[74] Once again it was demonstrated that the addition of $Ti(OiPr)_4$ did not improve the stereoselectivity of the product. Screening of various chiral ligands showed that the preformed titanium catalyst of (*R*)-H_8-BINOL (see Scheme 12) together with

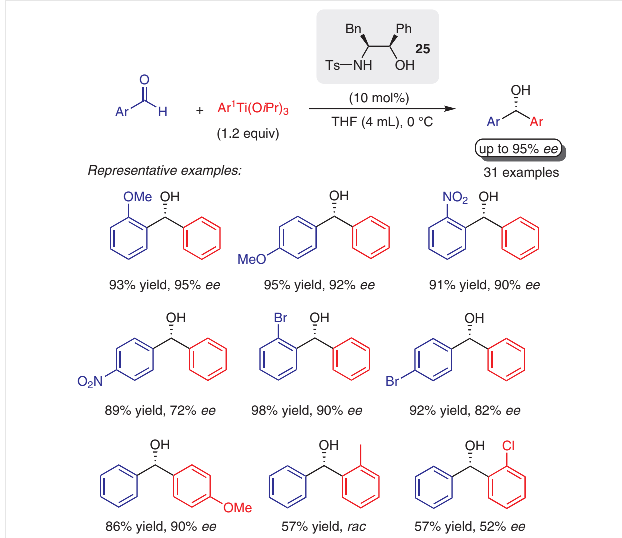
1.2 equivalents of ArTi(O*i*Pr)$_3$ afforded diarylmethanols with better *ee*. The optimized reaction conditions were employed to investigate the reaction scope. Arylation of several aromatic aldehydes afforded the desired diarylmethanols in excellent yields and high enantioselectivities (Scheme 27). With the intention to provide further insights into the reaction mechanism, the Gau group demonstrated from this study a positive linear effect, supporting a pathway in which the metallic active species involves only one (*R*)-H$_8$-BINOL in the transition state of the enantiofacial differentiation step. Autocatalysis was also evaluated, and the slower reaction and the linear effect of the catalytic system indicate that autocatalysis by the chiral alcohol products is negligible in comparison to the catalytic reaction.



Scheme 27 ArTi(O*i*Pr)$_3$ addition reactions of aryl aldehydes catalyzed by (*R*)-H$_8$-BINOL

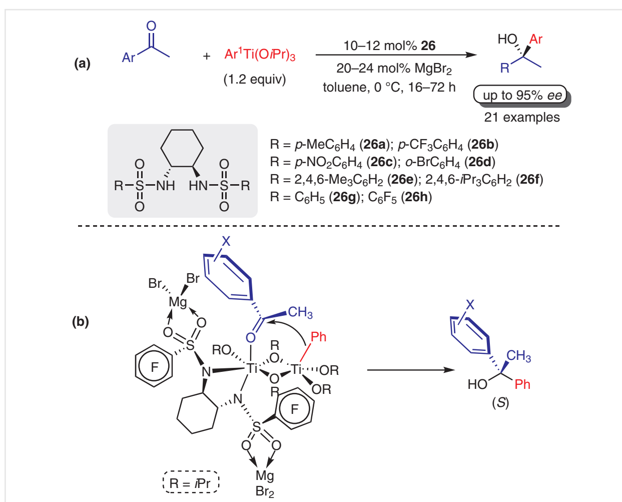
In subsequent work in 2015, Gau, Zhou, and co-workers reported the application of *N*-sulfonylated amino alcohol **25** in the enantioselective addition of ArTi(O*i*Pr)$_3$ to aldehydes.[75] Selected examples are shown in Scheme 28. A wide range of aldehydes were used and, in all cases the yields of the diarylmethanols were high. The enantiomeric excess, however, depended on the substitution pattern of the aromatic aldehydes. For instance, *para*-substituted benzaldehyde gave slightly lower enantioselectivities. The aryl transfer onto heteroaromatic 2-furaldehyde gave the product in good yield with 84% *ee*. The structure of the organotitanium reagents also tolerates some substitution, especially at the *para* position and the aryl group was transferred in high yields and with high *ee's*. *ortho*-Substituted arylorganotitanium reagents were problematic and *ee's* were generally lower.

Also in 2015, Gau, Zhou, and co-worker developed an asymmetric version of the magnesium-promoted enantioselective aryl addition of ArTi(O*i*Pr)$_3$ to ketones catalyzed by a titanium(IV) catalyst of *N*,*N*′-sulfonylated (1*R*,2*R*)-cyclohexane-1,2-diamines **26a–h** as the chiral ligand (Scheme



Scheme 28 Asymmetric ArTi(O*i*Pr)$_3$ addition to aryl aldehydes catalyzed by the titanium catalyst of an *N*-sulfonylated amino alcohol

29a).[76] Screening the catalytic properties of **26a–h** in asymmetric phenyl transfer from PhTi(O*i*Pr)$_3$ to 2′-acetonaphthone in the presence of inorganic salt $MgBr_2$ showed that the best result was obtained with 10 mol% of ligand **26h** in toluene at 0 °C. The effect of different magnesium halides was also evaluated and this showed that $MgBr_2$ and MgI_2 efficiently promoted the reactivity and enantioselectivity of the phenyl addition of PhTi(O*i*Pr)$_3$ to ketones. However, the use of $MgCl_2$ resulted in poor yield and very low enantioselectivity. It was essential to use two equivalents of MgX_2 relative to one molecule of the chiral ligand to achieve the best level of enantioselectivity. The scope of the reaction was examined and various *o*-, *m*-, and *p*-substituted

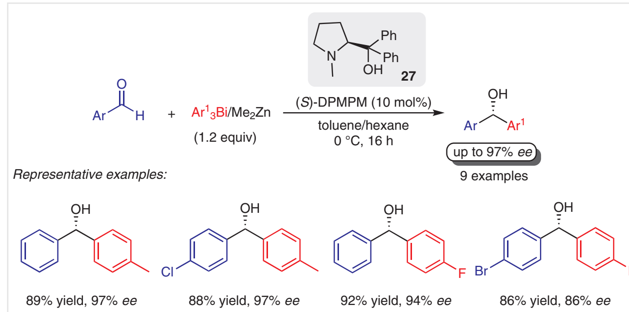


Scheme 29 (a) $MgBr_2$-promoted enantioselective aryl addition of ArTi(O*i*Pr)$_3$ to ketones catalyzed by a titanium(IV) catalyst of *N*,*N*′-sulfonylated (1*R*,2*R*)-cyclohexane-1,2-diamine. (b) Suggested stereochemical structure of the active titanium species.

acetophenones and heteroaryl methyl ketones reacted with generally phenyl, but also 4-substituted-phenyl and naphthyl, ArTi(O*i*Pr)$_3$ reagents to give the corresponding products in excellent yields and generally good to excellent enantioselectivity. It was suggested that the oxygen atoms of both sulfonyl groups coordinate with MgX_2 forming an active titanium species (Scheme 29b). The coordination of sulfonyl atoms to $MgBr_2$ or even the larger MgI_2 could provide a pocket of suitable size that allows insertion of the planar aromatic ring of the aromatic ketone for an addition of the phenyl nucleophile, producing the desired (*S*)-tertiary alcohol. When $MgCl_2$ was used a similar active species is obtained, however, due to the small size of $MgCl_2$ the titanium active species has a larger pocket size which also allows insertion of the methyl group affording the addition product in lower enantioselectivity.

8 Organobismuth Reagents

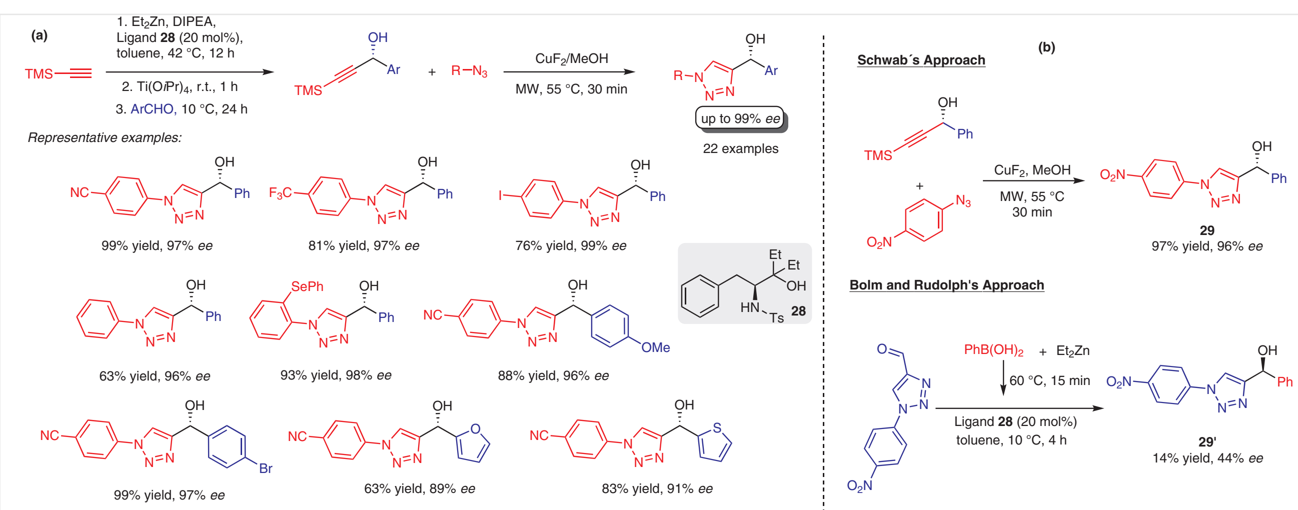
Organobismuth compounds have been used as catalysts and reagents in a plethora of reactions.[77] Taking this into account, in 2007, Sato and co-workers described for the first time the synthesis of diarylmethanols by enantioselective aryl addition using mixed triarylbismuthine and dialkylzinc reagents to aromatic aldehydes in the presence of a chiral β-amino alcohol (Scheme 30).[78] The best results were achieved with Me_2Zn, since the use of Et_2Zn resulted in the formation of a considerable amount of alkylated product. The organozinc reagent plays an important role since the reaction does not occur in its absence. (*S*)-Diphenyl(1-methylpyrrolidin-2-yl)methanol (DPMPM, **27**) (10 mol%) exhibited the best catalytic efficiency and provided the corresponding diarylmethanols in up to 93% yields and with up to 97% *ee*. From a mechanistic point of view, the active species resulting from the reaction using mixed triarylbismuthine and Me_2Zn is still unknown.



Scheme 30 Synthesis of chiral diarylmethanols by using mixed triarylbismuthine and Me_2Zn

9 Miscellaneous

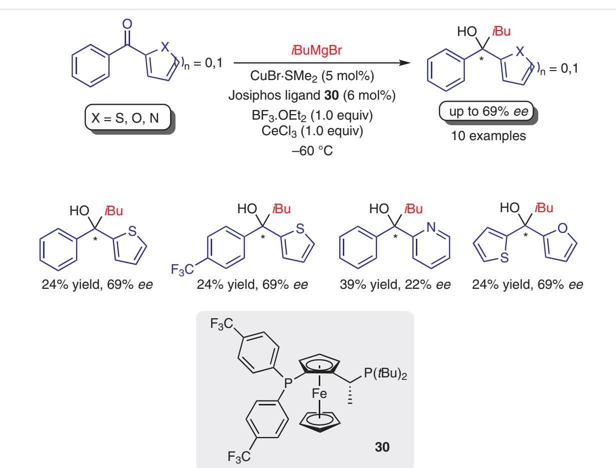
The synthesis of aryl(heteroaryl)methanols through the use of organozinc reagent is still a challenge since the traditional routes for the synthesis of these frameworks, which usually involve the enantioselective arylation of heteroaryl aldehydes are not so efficient. To overcome this issue, our group designed a straightforward sequential approach for the enantioselective synthesis of optically active 4-[aryl(hydroxy)methyl]-1,2,3-triazoles.[79] The approach is based on the enantioselective alkynylation of aldehydes in the presence of chiral ligand **28** followed by a one-pot, two-step desilylation/Cu(I)-catalyzed azide-alkyne 1,3-dipolar cycloaddition (CuAAC), providing the corresponding aryl(heteroaryl)methanols and diheteroarylmethanols in excellent yields and with high levels of enantioselectivity (Scheme 31a). To further demonstrate the efficiency of our



Scheme 31 (a) Sequential approach for the enantioselective synthesis of optically active 4-[aryl(hydroxy)methyl]-1,2,3-triazoles. (b) Comparison of our approach to known work.

methodology, we used a conventional arylation protocol to prepare compound **29** and compared the outcomes (Scheme 31b). In comparison with our protocol, the enantioselective arylation using the approach of Bolm and Rudolph[22] afforded the final product **29′** in 14% yield with 44% *ee*. This result shows that this approach, though very versatile, does not apply to more complex substrates such as aldehydes bearing 1,2,3-triazoles, a gap that was successfully filled by our protocol.

The asymmetric synthesis of tertiary diarylalcohols and aryl(heteroaryl)alcohols still remains a great challenge, especially when the corresponding ketones with two aryl substituents were used as starting materials. This remarkable difference in reactivity of diaryl ketones compared to aryl aldehydes or even aryl alkyl ketones is believed to be the result of the negligible steric and electronic differences between the two aryl substituents on the carbonyl moiety. Taking this into account, in 2015 Harutyunyan and coworkers reported a new strategy to access chiral tertiary diarylalcohols through asymmetric copper-catalyzed direct alkylation of aryl heteroaryl and diheteroaryl ketones by using Grignard reagents in the presence of Joshiphos ligand **30** (Scheme 32).[80] However, under these reaction conditions, the reaction led to the formation of racemic secondary alcohols by non-catalytic Meerwein–Ponndorf–Verley (MPV)-type reduction. Further refinements on the reaction conditions were made and it was found that the addition of a mixture of $BF_3{\cdot}OEt_2/CeCl_3$ (1:1) resulted in an improvement of the reactivity of the substrates and avoided the MPV-type reduction. The scope of the alkylation was examined using other aryl heteroaryl and diheteroaryl ketones and gave the corresponding 1-aryl-1-(heteroaryl)alkan-1-ols in 6–69% *ee*.



Scheme 32 Catalytic asymmetric alkylation of aryl heteroaryl ketones

10 Conclusion

The field of enantioselective arylation and heteroarylation of aldehydes and ketones has seen a flurry of activity over the past decades, and huge advances have been made in both the development of novel aryl- and heteroarylmetal reagents and the mechanistic study of these transformations. These advances provide a straightforward strategy for the construction of optically active diaryl, aryl heteroaryl, and diheteroaryl alcohols, which are important building blocks for pharmacologically and biologically active compounds. Although several organometallics were readily available, for many years arylzincs and arylboronic acids were almost exclusively and widely used as the suitable source of the aryl group for the asymmetric synthesis of chiral diaryl alcohols. The great success associated with these two approaches could be explained based on the higher reactivity of commercially available and cost-effective aryl Grignard and aryllithium reagents, which, has limited their application in catalytic asymmetric processes. The advent of new chiral catalytic systems and synthetic procedures for the enantioselective arylation and heteroarylation of carbonyl compounds creates new opportunities to increase the scope of organometallic reagents used for the synthesis of chiral secondary and tertiary alcohols. The literature surveyed in this review has shown that a diverse range of organometallic reagents such as organolithium, Grignard, organoaluminum, organotitanium, and organobismuth reagents can be now efficiently used to access the chiral diaryl, aryl heteroaryl, and diheteroaryl alcohols in high yields and with high enantiomeric ratios. Despite the impressive advances described in this review, the future seems similarly bright as there are many challenges that remain to be addressed. The first challenge is the high reactivity of aryllithium and aryl Grignard reagents, most of the procedures require the use of super-stoichiometric amounts of $Ti(OiPr)_4$ to prepare the arylation reagents by transmetalation. The second challenge, the slow addition of the arylation reagents to carbonyl compounds over a long period of time, or the change of the solvent during the reaction makes the process less practical. Concerning these drawbacks, it is highly required that the new methodologies are more economically efficient thus allowing the use of substoichiometric amounts of titanium sources. A further effort is also needed to improve the enantioselective arylation of ketones, especially when diaryl ketones are used as an electrophile. Thus, there is no doubt that further progress will open for new synthetic routes and opportunities for organic chemists.

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